INTEGRATION

Defn: (Integration)

Integration or anti-derivative of a continuous function f(x), is the reverse process of differentiation.

* Indefinite integration: If you integrate f(x) without limit interval on x-values, we obtain an indefinite integral.

$$\int f(x)\,dx = \underbrace{F(x) + C}_{\text{Output is function plus a constant.}}, \quad \text{where: } F'(x) = \frac{d}{dx}F(x) = f(x).$$

C - Abitrary Constant.

** Definite Integral: If you integrate f(x) with limit interval on x, say $a \le x \le b$, we obtain a definite integral

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]\Big|_{x=a}^{x=b} = \underbrace{F(b) - F(a)}_{\text{Output is a real value.}}, \quad \text{where } F'(x) = f(x)$$

INTEGRALS OF STANDARD FUNCTIONS

1.) $\frac{d}{dx}(x^n) = nx^{n-1}$ $\quad\therefore \int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \ne -1)$

2.) $\frac{d}{dx}(\ln x) = \frac{1}{x}$ $\quad\therefore \int \frac{1}{x}\,dx = \ln x + C \quad (n = -1)$

3.) $\frac{d}{dx}(e^{ax}) = ae^{ax}$ $\quad\therefore \int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$

4.) $\frac{d}{dx}(a^x) = a^x \ln a$ $\quad\therefore \int a^x\,dx = \frac{1}{\ln a}a^x + C$

5) $\frac{d}{dx}(\cos ax) = -a\sin ax$ $\quad\therefore \int \sin ax\,dx = \frac{-1}{a}\cos ax + C$

6) $\frac{d}{dx}(\sin ax) = a\cos ax$ $\therefore \int \cos ax\,dx = \frac{1}{a}\sin ax + C$

7) $\frac{d}{dx}(\tan ax) = a\sec^2 ax$ $\therefore \int \sec^2 ax\,dx = \frac{1}{a}\tan ax + C$

HYPERBOLIC FUNCTIONS

* 8) $\frac{d}{dx}(\cosh ax) = \sinh ax$ ✓* $\therefore \int \sinh ax\,dx = \frac{1}{a}\cosh ax + C$ ✓*

** 9) $\frac{d}{dx}(\sinh ax) = \cosh ax$ ✓** $\therefore \int \cosh ax\,dx = \frac{1}{a}\sinh ax + C$ ✓**

*** 10) $\frac{d}{dx}(\tanh ax) = a\,\mathrm{sech}^2 ax$ ✓*** $\therefore \int \mathrm{sech}^2 ax\,dx = \frac{1}{a}\tanh ax + C$ ✓***

11) $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$ $\therefore \int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}x + C$

12) $\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$ $\therefore \int \frac{-dx}{\sqrt{1-x^2}} = \cos^{-1}x + C$

13) $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$ $\therefore \int \frac{1}{1+x^2}\,dx = \tan^{-1}x + C$

* HYPERBOLIC INVERSE FUNCTIONS

* 14) $\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{x^2+1}}$ $\therefore \int \frac{1}{\sqrt{x^2+1}}\,dx = \sinh^{-1}x + C$

* 15) $\frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}}$ $\therefore \int \frac{1}{\sqrt{x^2-1}}\,dx = \cosh^{-1}x + C$

* 16) $\frac{d}{dx}(\tanh^{-1}x) = \frac{1}{1-x^2}$ $\therefore \int \frac{1}{1-x^2}\,dx = \tanh^{-1}x + C$

Example 1

Evaluate the following integrals:

(a) $\int \left(4x^6 - \frac{3}{\sqrt{x}} + e^{-3x}\right) dx$

(b) $\int \left(5^x + \frac{4}{1+x^2} - \frac{7}{x}\right) dx$

(c) $\int \left(x^{\frac{1}{3}} + \frac{1}{\sqrt{x^2+1}} - \sin 2x\right) dx$

(d) $\int \left[5\sqrt{x} + \sec^2 5x - 5\cos 3x\right] dx$

Solution

(a) $\int \left(4x^6 - 3x^{-1/2} + e^{-3x}\right) dx = \frac{4x^7}{7} - \frac{3x^{1/2}}{1/2} + \frac{1}{-3} e^{-3x} + C$

$$= \frac{4}{7}x^7 - 6\sqrt{x} - \frac{1}{3}e^{-3x} + C$$

(b) $\int \left(5^x + \frac{4}{1+x^2} - \frac{7}{x}\right) dx = \int 5^x dx + 4\int \frac{1}{1+x^2} dx - 7\int \frac{1}{x} dx$

$$= \frac{1}{\ln 5} 5^x + 4\tan^{-1} x - 7\ln x + C$$

(c) $\int \left(x^{1/3} + \frac{1}{\sqrt{x^2+1}} - \sin 2x\right) dx = \int x^{1/3} dx + \int \frac{1}{\sqrt{x^2+1}} dx - \int \sin 2x \, dx$

$$= \frac{x^{4/3}}{4/3} + \sinh^{-1} x + \frac{1}{2}\cos 2x + C$$

$$= \frac{3}{4}x^{4/3} + \sinh^{-1} x + \frac{1}{2}\cos 2x + C$$

(d) $\int \left(5x^{1/2} + \sec^2 5x - 5\cos 3x\right) dx = 5\int x^{1/2} dx + \int \sec^2 5x \, dx - 5\int \cos 3x \, dx$

$$= \frac{5x^{3/2}}{3/2} + \frac{1}{5}\tan 5x - \frac{5}{3}\sin 3x + C$$

$$= \frac{10}{3}x^{3/2} + \frac{1}{5}\tan 5x - \frac{5}{3}\sin 3x + C$$

TECHNIQUES OF INTEGRATION

① FUNCTION SUBSTITUTION

If $\int f[g(x)]\,dx$ then let $u = g(x)$, $du = g'(x)\,dx$. Substitute in the integral.

$$\boxed{\int \frac{f'(x)}{f(x)}\,dx = \ln[f(x)] + C \qquad \text{Let } u = f(x)\ ;\ du = f'(x)\,dx}$$

$$\boxed{\int f'(x)\,f(x)\,dx = \frac{1}{2}[f(x)]^2 + C \qquad \text{Let } u = f(x)\ ,\ du = f'(x)\,dx}$$

Example 2

Evaluate the following integrals:

(a) $\int x(3x^2+4)^6\,dx$ (b) $\int e^{3x-2}\,dx$ (c) $\int (6x+1)e^{3x^2+x-1}\,dx$

Solution

(a) $\int x(3x^2+4)^6\,dx = \int u^6 \cdot \frac{1}{6}\,du = \frac{1}{6}\cdot\frac{u^7}{7} + C = \frac{1}{42}u^7 + C$

Let $u = 3x^2+4$

$du = 6x\,dx$

$\frac{1}{6}du = x\,dx$

$= \frac{1}{42}(3x^2+4)^7 + C$

(b) $\int e^{3x-2}\,dx = \int e^u \frac{1}{3}\,du = \frac{1}{3}e^u + C = \frac{1}{3}e^{3x-2} + C$

Let $u = 3x-2$

$du = 3\,dx \Rightarrow \frac{1}{3}du = dx$

(c) $\int (6x+1)e^{3x^2+x-1}\,dx = \int e^u\,du = e^u + C = e^{3x^2+x-1} + C$

$u = 3x^2+x-1$

$du = (6x+1)\,dx$

Example 3

Evaluate the following integrals:

(a) $\int \frac{6x-1}{3x^2-x+5}\,dx$ (b) $\int \frac{4x+2}{2x^2+2x-1}\,dx$

(c) $\int \frac{x-1}{x^2-2x+3}\,dx$ (d) $\int \frac{\sin x}{1-\cos x}\,dx$.

Solution

(a) $\int \frac{6x-1}{3x^2-x+5}\,dx = \int \frac{1}{u}\,du = \ln u + C = \underline{\ln(3x^2-x+5)+C}$

$u = 3x^2-x+5$

$du = (6x-1)\,dx$

(b) $\int \frac{4x+2}{2x^2+2x-1}\,dx = \int \frac{du}{u} = \ln u + C = \underline{\ln(2x^2+2x-1)+C}$

Let $u = 2x^2+2x-1$

$du = (4x+2)\,dx$

(c) $\int \frac{x-1}{x^2-2x+3}\,dx = \frac{1}{2}\int \frac{2(x-1)}{x^2-2x+3}\,dx = \underline{\frac{1}{2}\ln(x^2-2x+3)+C}$

Let $u = x^2-2x+3$

$du = (2x-2)\,dx$

(d) $\int \frac{\sin x}{1-\cos x}\,dx = \int \frac{1}{u}\,du = \ln u + C = \underline{\ln(1-\cos x)+C}$

Let $u = 1-\cos x$

$du = \sin x$

INTEGRATION BY PARTS

Given $\int uv'\,dx = uv - \int u'v\,dx$

Example 1

Integrate (a) $\int xe^{2x}\,dx$ (b) $\int x^2 \sin x\,dx$

(c) $\int e^{2x} \cos x\,dx$

Solution

(a) $\int xe^{2x}\,dx = \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}\,dx = \underline{\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C}$

Let $u = x$ $v' = e^{2x}$

$u' = 1$ $v = \frac{1}{2}e^{2x}$

(b) $\int \underbrace{x^2}_{u} \underbrace{\sin x}_{v'}\,dx$, let $u = x^2$; $v' = \sin x$

$u' = 2x$; $v = -\cos x$

$= -x^2\cos x + 2\int \underbrace{x}_{u} \underbrace{\cos x}_{v'}\,dx$ Apply Integration by parts again

$u = x$ $v' = \cos x$

$u' = 1$ $v = \sin x$

$= -x^2\cos x + 2\left[x\sin x - \int \sin d x\right]$

$= \underline{-x^2\cos x + 2x\sin x + 2\cos x + C}$

(c) $\int e^{2x}\cos x\,dx = e^{2x}\sin x - 2\int e^{2x}\sin x\,dx$

Let $u = e^{2x}$ $v' = \cos x$ | Let $u = e^{2x}$ $v' = \sin x$ "Apply Int. by parts again."

$u' = 2e^{2x}$ $v = \sin x$ | $u' = 2e^{2x}$ $v = -\cos x$

So that

$$\int e^{2x}\cos x\,dx = e^{2x}\sin x - 2\left[-e^{2x}\cos x + 2\int e^{2x}\cos x\,dx\right]$$

$$\int e^{2x}\cos x\,dx = e^{2x}\sin x + 2e^{2x}\cos x - 4\int e^{2x}\cos x\,dx$$

$$5\int e^{2x}\cos x\,dx = e^{2x}\sin x + 2e^{2x}\cos x$$

$$\therefore \int e^{2x}\cos x\,dx = \underline{\frac{1}{5}\left(e^{2x}\sin x + 2e^{2x}\cos x\right)}$$